
Wave and Bottom Sediment Interactions over a Submerged Sand Bank during the Winter Cold-Front Season, Western Louisiana

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ABSTRACT

The Louisiana coast is experiencing severe coastal land losses due to geological processes and human intervention. Replenishing the eroded beaches and the barrier islands with sand from offshore borrow sites is a plausible way to restore them. To assess the potential effects of sand mining from these offshore banks, in terms of physical processes, a field survey was conducted on March 2004 at Sabine Bank, 15 km offshore in western Louisiana. Some of the data from the survey are presented here. The results show that waves strongly affect the bottom sediment re-suspension rather than currents, during cold front and post-frontal phases that occurred during the deployment. Shear velocity and turbidity have a positive correlation, although the linear coefficient of determination (R^2) was 0.43. Maximum normalized cross-correlation coefficient between the two parameters was 0.82. The above results suggest that the waves greatly affect the bottom sediment and both parameters have some degree of positive correlation, which is not necessarily linear. The non-linear correlation may be associated with bottom roughness and sediment type as well as wave parameters. A simplification of the relationship using a stability parameter is also examined.

INTRODUCTION

Louisiana's coast has unique characteristics in terms of a low energy wave environment, high land subsidence, and frequent cold fronts in winter and hurricanes in summer (Wright *et al.*, 1997; Friedrichs *et al.*, 2000; Stone, 2000; Stone *et al.*, 2004). Historical research shows that along the Louisiana coast, waves are too weak to re-suspend sediment during fair weather; during cold fronts sediment re-suspension can occur (Wright *et al.*, 1997; Friedrichs *et al.*, 2000).

Over the decades, the Louisiana coast has been undergoing severe coastal land loss due to sediment compaction, sediment deficit from the Mississippi River and its distributaries, wave/wind erosion, and hurricane overwash (Penland *et al.*, 1988; Roberts, 1997). One of the possible solutions to prevent coastal land loss is to supply sand from potential offshore sites (Drucker *et al.*, 2004; Stone *et al.*, 2004). Sabine Bank, which is a remnant of palaeo-beach ridges, submerged due to rapid sea-level rise during the late Holocene, is thought to be one of the potential offshore sand borrow sites (Morton and Gibeaut, 1995; Drucker *et al.*, 2004; Rodriguez *et al.*, 2004). However, submerged sand banks are thought to reduce the energy of approaching waves and storm surges, and also serve as a nursery for benthic habitats (Stone *et al.*, 2004; U.S. Geological Survey, 2003). Therefore, to as-

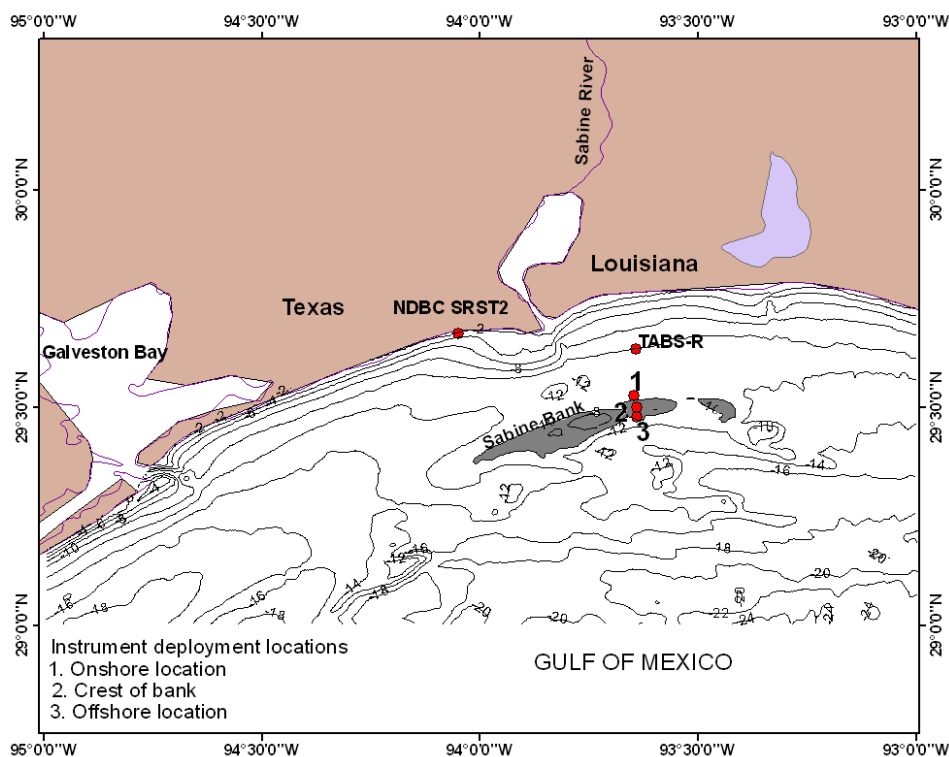


Figure 1. Map of Sabine Bank and our observation stations.

sess the potential effects of sand mining, in terms of physical processes, a field survey of waves, currents and sediment transport was undertaken for a period of 32 days in March, 2004 at Sabine Bank, to determine wave-bottom sediment interactions, which had not previously been undertaken in this area (Fig. 1).

The objective of this paper is to present the characteristics of waves and bottom sediment interactions over a shallow sand bank, Sabine Bank, based on the field survey.

THREE DISTINCT METEOROLOGICAL PATTERNS

Meteorological data obtained from a nearby National Data Buoy Center (NDBC) buoy station, SRST2, show that there appear to be three distinct meteorological patterns: fair weather, cold fronts and post-frontal phases (Kobashi *et al.*, 2005). Between October and May, frequent cold-front passages (*i.e.*, 20-30 times per season) occur along the northern Gulf of Mexico (Roberts *et al.*, 1989). Typically winds blow from southern quadrants during fair weather, but once cold fronts pass the area, wind direction dramatically changes from south to north, wind speed increases, and air temperature drops (Stone, 2000). During our survey period, 11 cold fronts passed the survey area.

SIGNIFICANCE OF WAVES ON BOTTOM SEDIMENTS DURING COLD FRONTS AND POST-FRONTAL PHASES

Time series of winds, waves, currents, optical backscatter sensor (OBS) data representing turbidity, shear velocity and shear stress due to waves are given in Figure 2. In general, waves are strongly associated with the local wind field. Variance of significant wave height (H_s) was high during high-wind events such as cold fronts

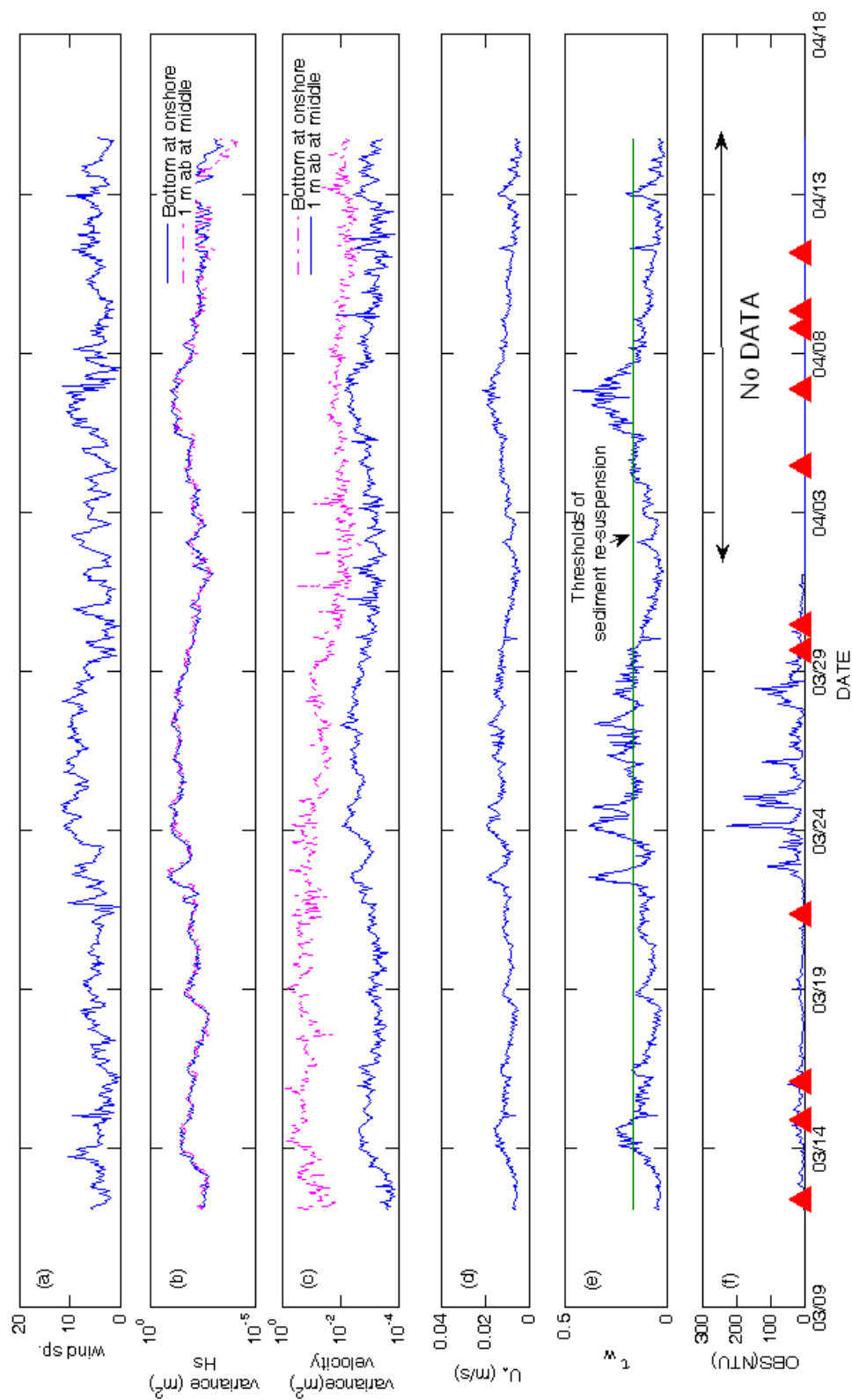


Figure 2. Time series plots of (a) wind, (b) variance of H_s , (c) variance of current speed, (d) shear velocity (U_*), (e) shear stress (τ_w), and (f) turbidity (OBS). Second and third figures show the variances at the middle station (dash-dotted line) and at the onshore station (single line). Red shaded triangles show the passages of cold fronts. The line parallel to x-axis at the fifth figure shows the threshold of sediment re-suspension based on Shields parameters.

(red shaded triangles in Figure 2) and post-frontal phases (double arrow in Figure 2) at both onshore and middle stations (Kobashi *et al.*, 2005). H_s was also correlated with turbidity (OBS) data and consequently shear-stress values that exceed the threshold of sediment re-suspension were also associated with high turbidity. Both bottom currents at the onshore station and currents at 1 m above the bottom (ab) at the middle station were associated with winds and waves; however, the relationship between winds and currents at the middle station was not as clear as that at the onshore station. Shear stress due to waves is associated with the three meteorological conditions. Shear stress values exceeded the threshold of sediment re-suspension during cold fronts and post-frontal phases and was lower than the threshold during fair weather. However, it is noted that not all of the cold fronts cause higher shear stress than the threshold, but that depends on the intensity and the duration of the storms.

WAVES AND BOTTOM SEDIMENT INTERACTIONS

In Figure 3, a time series of shear velocity and OBS data obtained during cold fronts and post-frontal phases are presented. These two parameters are strongly correlated; high turbidity occurs during the period of high shear velocity. However, there exist time lags between these two variations (see Sheremet *et al.*, 2005). The correlation between turbidity and shear velocity is shown in Figure 4 (left). It is apparent that there is a positive correlation between the two, although not linear. A coefficient of determination of 0.43 was calculated. The maximum normalized cross-correlation coefficient is 0.82. Sediment concentration is thought to be related to sediment type, bottom bed forms as well as wave-induced shear velocity. Vincent and Green (1990) proposed that sediment re-suspension is associated with wave-induced thresholds and vortex shedding from a rippled bed. Glenn and Grant (1987) show a detailed theoretical approach of sediment concentration and waves. The relationship can be characterized using a stability parameter (z/L), and the parameter can be described as the sum of the sediment concentration and sediment parameters for each sediment class at each depth level as (Glenn and Grant, 1987; Vincent and Green, 1990).

$$\frac{z}{L} = \frac{\kappa}{U_*^3} \sum_n g(S_n - 1) * w_{f_n} * C_n(z) \dots\dots\dots (1)$$

where z/L is stability parameter, L is Monin-Obukhov length, κ is von-Karman constant ($= 0.4$), g is gravitational acceleration (9.8 m/sec^2), U_* is shear velocity, S_n is relative density of sediment in class n , w_{f_n} is settling velocity of the sediment in class n , and $C(z)$ is sediment concentration (SSC) at a certain depth (z). Here, κ , g , S , and w_f are time-independent and known values for each class of sediment. Turbidity and SSC have a positive linear correlation (Sheremet *et al.*, 2005). In addition, in the atmospheric surface layer, the Monin-Obukhov length is often assumed to be constant (Glenn and Grant, 1987). Therefore, if these conditions can be applied and values for each sediment class at each depth level are known, Equation 1 can be transformed as follows.

$$OBS = \sum_n a_n * U_*^3 \dots\dots\dots (2)$$

$$\sum_n a_n = \frac{1}{\kappa \cdot \sum_n g(S_n - 1) \cdot w_{f_n}} \cdot \frac{z}{L} \dots\dots\dots (3)$$

where a_n is a known value for each sediment class at each depth level. Equation 2 shows that the OBS value can be represented as a function of a_n and the shear velocity. Figure 4 (right) shows the relationship between the OBS and the shear velocity including parabolas with different arbitrary a_n values. From Equation 3, higher “ a ” values suggest lower sediment density (S), lower settling velocity (w_f), and high stability parameter (z/L). Also, higher “ a ” values show higher turbidity for the same shear velocity (Fig. 4, right). Glenn and Grant (1987) noted that higher settling velocity has lower sediment concentration and vice versa. From Equations 2 and 3, higher “ a ” values suggest low settling velocity and have high turbidity which is proportional to SSC. Therefore, our hypothesis follows the interpretation of Glenn and Grant (1987). It should be reiterated, however, that these data and analyses are preliminary and serve as a precursor for more detailed future work.

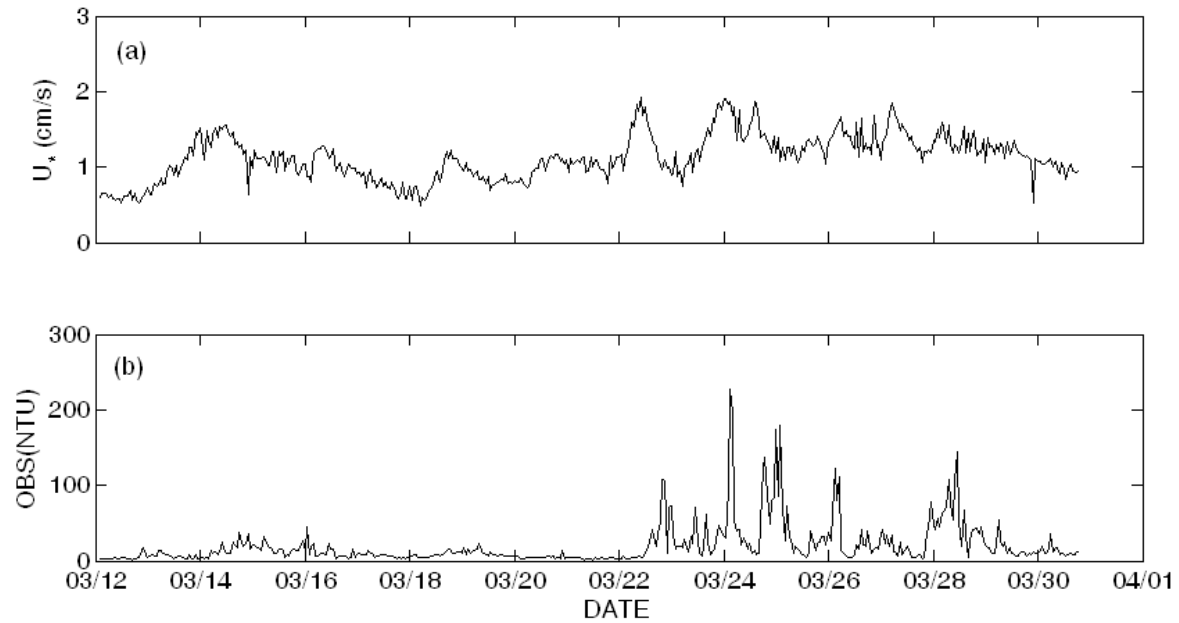


Figure 3. Time series plots of (a) shear velocity (U_*) and (b) turbidity (OBS).

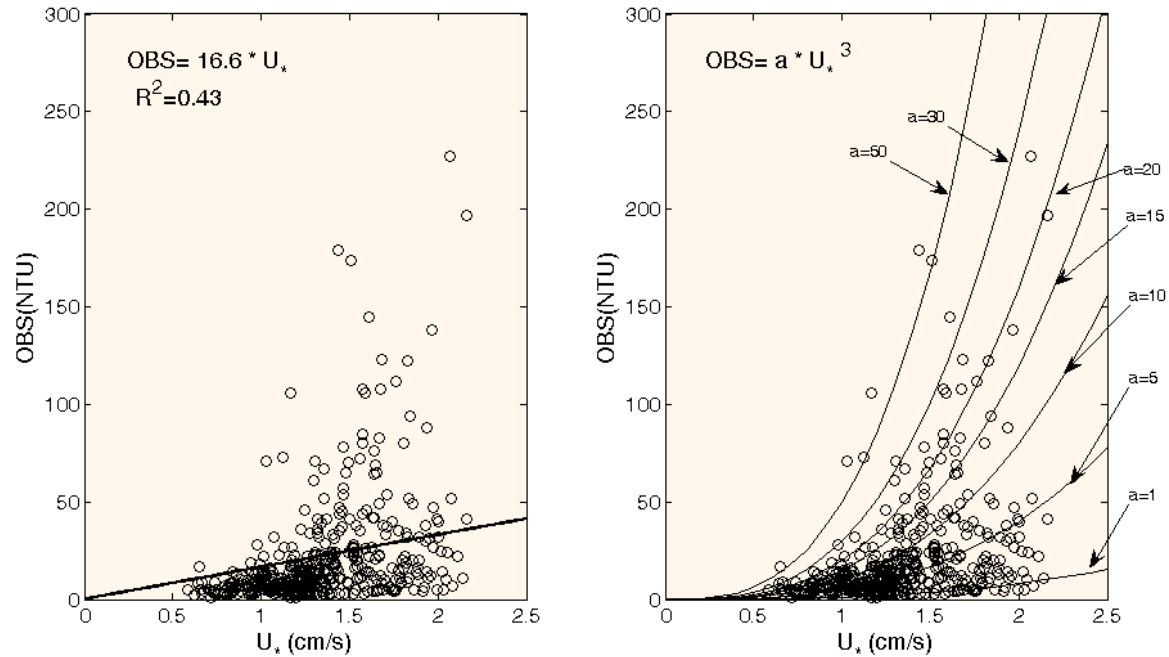


Figure 4. The relationship between shear velocity (U_*) and turbidity (OBS) with linear regression line (left) and hypothetical parabolas with arbitrary "a" values (right).

CONCLUSIONS

The following conclusions can be drawn from the above described analysis:

Significant wave height is correlated with turbidity (OBS) depending on the intensity and the duration of the storms.

Shear stress due to waves was higher than the threshold for sediment re-suspension during cold fronts and post-frontal phases.

Shear velocity and turbidity have a positive correlation although the linear regression coefficient of determination (R^2) was 0.42; the maximum normalized cross-correlation coefficient was 0.82.

The above results suggest that waves during severe weather conditions are more effective in sediment re-suspension than currents. Also, sediment re-suspension is likely to occur only during cold fronts and post-frontal phases during our survey period. The correlations between shear velocity and turbidity are not linear and SSC which is proportional to turbidity is inferred to be proportional to a parabola once other parameters which can be known for each sediment class and each depth level are determined.

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